ELEC3104: Digital Signal Processing Lab

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Topic 2 – Convolution

Exercise: Given $x_1\,[n] = \delta[n] + 2\,\delta[n-1] + 3\delta[n-2] + 4\delta[n-3]$

$$x_2\,[n] = \delta[n-1] - 2\,\delta[n-2]$$

a. Find $x_1[n]*x_2[n]$
b. Use MATLAB command "conv" to check the results.

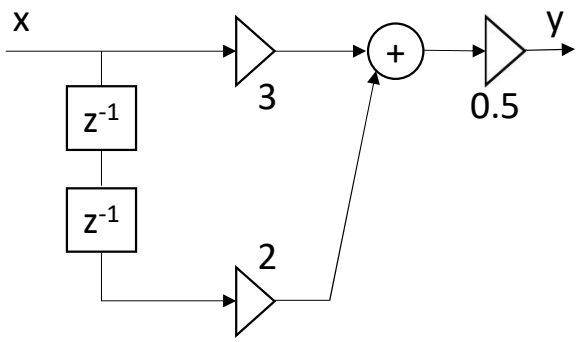
Exercise: In MATLAB, write a function *my_conv* that takes two inputs x1, x2 and produces the output y equal to the convolution of x1 and x2. Compare the result produced by your *my_conv* function and the MATLAB in-built function *conv*

Topic 2 – Discrete Time System Implementation

Exercise: Given a system described by the following block diagram. In MATLAB, compute its output y when the input signal is `x = [1 -3 2 4 -2 2 6 2 4 -2]` using three different methods as given below:

a. Using a for loop
b. 'conv' command
c. 'filter' command

Compare the results of (a), (b) and (c)



Topic 2 – Pole-Zero Description of Discrete-Time Systems

Exercise: A discrete-time system described by its transfer function is shown below:

$$H(z) = \frac{z^8 - (0.95)^8}{z^7(z - 0.95)}$$

Using MATLAB,

a. Find all the poles and zeros of the system and plot them in the z-plane
b. Compute and plot the magnitude response of the system
c. Compute and plot the impulse response of the system

Hint: The in-built functions *freqz* and *impz* could be used to compute the frequency and impulse responses

Topic 2 – Pole-Zero Description of Discrete-Time Systems (2)

Exercise: A 2nd order system has two zeros at the origin and two poles as $p_1 = re^{j\pi/4}$, $p_2 = re^{-j\pi/4}$, described by the following transfer function

$$H(z) = \frac{z^2}{(z - p_1)(z - p_2)}$$

a. Using MATLAB, compute and plot the magnitude response, impulse response, and pole-zero plane (z-plane) of three different 2nd order systems.
   System 1: r = 0.8
   System 2: r = 0.9
   System 3: r = 0.99

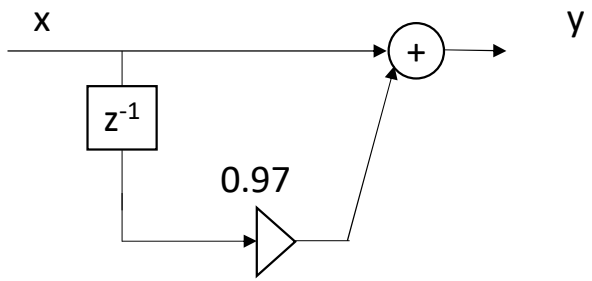
b. Comment on the impact of the location of poles to the impulse response and magnitude response for three systems of (a)

5

Topic 2 z-Transform, Impulse Response and Frequency Response

Exercise: There are two different methods to compute the frequency response of a discrete-time system. The first method is using the z-transform and the second one is using Discrete Fourier Transform (DFT) of its impulse response

In MATLAB, write a program to compute the frequency response of the following system and compare the magnitude and phase responses obtained by the above two methods .



Topic 2 – Discrete Fourier Transform (DFT)

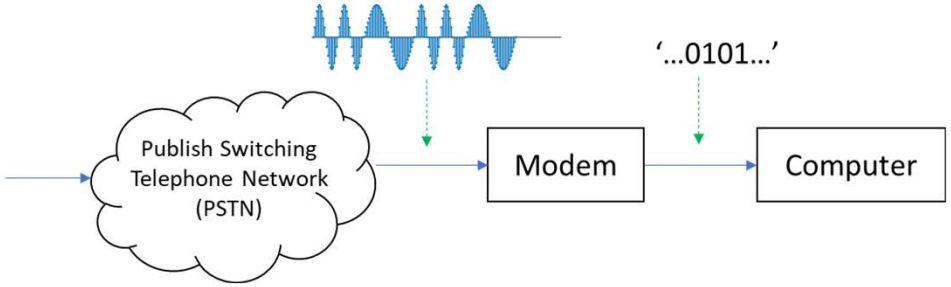
Exercise: Given a discrete time sinusoid signal saved in a wav file '*signal1.wav*' located in Moodle folder Additional Resources/ Wav files for DSP Labs. Using MATLAB,

a. Write a function *frequency_estimation* to read this wav file, automatically estimate the frequency of the sinusoid signal.
b. How accurate your frequency estimation? Discuss

Hint: The in-built functions *audioread* could be used to read the wav file and signal sampling rate.

Topic 2 – Discrete Fourier Transform (DFT)

Exercise: In ancient times, modems worked by transmitting digital data through telephone lines. Specifically, during transmission modems converted a stream of 1's and 0's into an audio signal where 1's were represented by a particular frequency and 0's by another frequency. A modem acting as a receiver would convert the audio signal back into a stream of 1's and 0's. You task today is to design and implement a system in MATLAB (as a MATLAB function) that will accept such audio signals (supplied to you as wav files – 'bit_sequence1.wav', 'bit_sequence2.wav', 'bit_sequence3.wav' and 'bit_sequence4.wav' located in Moodle folder Additional Resources/ Wav files for DSP Labs) as input and convert them into sequences of 1's and 0's accurately.



File	Decoded Bit Sequence
bit_sequence1.wav	
bit_sequence2.wav	
bit_sequence3.wav	
bit_sequence4.wav	

You have been given the following information about the audio signal:

1. Each symbol (i.e., 0 or 1) is represented by a sinusoid of a single frequency.
2. Each transmission begins with 0101.
3. The transmission rate is 20 bits/second